$\times$ [USD2.45 − $E$(EPS|*declining interest rate environment*)]$^2$

= 0.25(USD2.60 − USD2.4875)$^2$ + 0.75(USD2.45 − USD2.4875)$^2$ = 0.004219

Similarly, $\sigma^2$(EPS | *stable interest rate environment*) is found to be equal to

= 0.60(USD2.20 − USD2.12)$^2$ + 0.40(USD2.00 − USD2.12)$^2$ = 0.0096

These are **conditional variances**, the variance of EPS given a *declining interest rate environment* and the variance of EPS given a *stable interest rate environment*. The relationship between unconditional variance and conditional variance is a relatively advanced topic. The main points are that (1) variance, like expected value, has a conditional counterpart to the unconditional concept; and (2) we can use conditional variance to assess risk given a particular scenario.

**EXAMPLE 3**

## BankCorp's Earnings per Share, Part 3

Continuing with the BankCorp example, you focus now on BankCorp's cost structure. One model, a simple linear regression model, you are researching for BankCorp's operating costs is

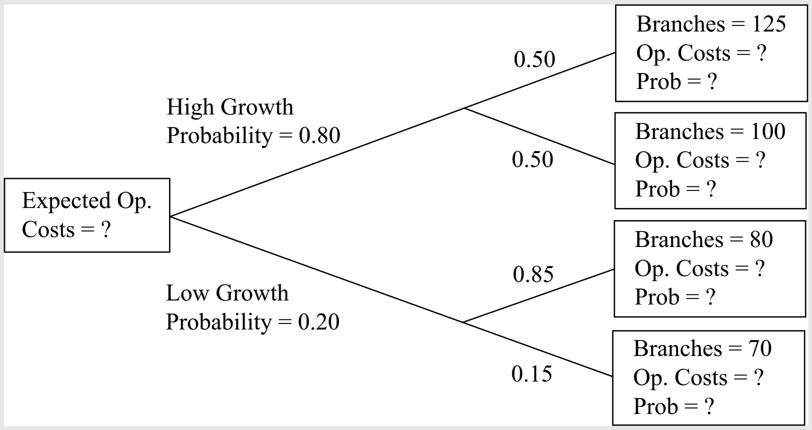
$$\hat{Y} = a + bX,$$

where $\hat{Y}$ is a forecast of operating costs in millions of US dollars and $X$ is the number of branch offices; and $\hat{Y}$ represents the expected value of $Y$ given $X$, or $E(Y \mid X)$. You interpret the intercept $a$ as fixed costs and $b$ as variable costs. You estimate the equation as follows:

$$\hat{Y} = 12.5 + 0.65X.$$

BankCorp currently has 66 branch offices, and the equation estimates operating costs as 12.5 + 0.65(66) = USD55.4 million. You have two scenarios for growth, pictured in the tree diagram in Exhibit 3.

**Exhibit 3: BankCorp's Forecasted Operating Costs**




1. Compute the forecasted operating costs given the different levels of operating costs, using $\hat{Y} = 12.5 + 0.65X$. State the probability of each level of the number of branch offices. These are the answers to the questions in the terminal boxes of the tree diagram.

   **Solution:**

   Using $\hat{Y} = 12.5 + 0.65X$, from top to bottom, we have